MILLING, SHAPER AND PLANER

Mechanisms • Indexing • Operations • Cutters • Machining Parameters • Machining Time

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1. Milling Machine Mechanism

A milling machine removes material by means of a rotating multi-point cutter. The cutter provides the primary cutting motion, while the workpiece is given controlled feed motion by the table. Depending on machine type, the spindle may be horizontal or vertical.

1.1 Principal motions

Motion	Description
Cutting motion	Rotation of the milling cutter about the spindle axis.
Longitudinal feed	Table movement parallel to its length.
Cross feed	Saddle/table movement perpendicular to longitudinal feed.
Vertical feed	Knee or spindle movement to set the depth of cut.

1.2 Drive mechanism

- Power from the motor is transmitted to the spindle through belts, gears or a geared speed box.
- Spindle-speed selection provides different cutter speeds to suit cutter diameter and work material.
- Feed motion is transmitted through the feed gearbox to the lead screws of the table, saddle and knee.
- In a horizontal milling machine the cutter is commonly mounted on an arbor supported by an arbor support.
- In a vertical milling machine the cutter is held directly in the vertical spindle using a collet, chuck or arbor.

1.3 Main functional units

Unit	Function
Base and column	Provide rigidity and support; column houses spindle drive.
Knee	Moves vertically and supports saddle/table.
Saddle	Provides cross movement.
Table	Carries the work and gives longitudinal feed.
Spindle	Rotates the cutter.
Arbor / cutter holder	Supports and drives the cutter.
Feed gearbox	Provides a range of table-feed rates.
Overarm and arbor support	Support the arbor in horizontal milling.

2. Milling Process

During milling, each tooth of the rotating cutter enters the work, removes a chip, and leaves the work. Since successive teeth cut intermittently, the cutter is subjected to repeated loading. Feed per tooth and cutting speed are therefore important process variables.

2.1 Up milling (conventional milling)

- Cutter rotates against the direction of work feed at the point of contact.
- Chip thickness starts approximately from zero and increases to a maximum.
- Tendency is to lift the workpiece from the table; rigid clamping is necessary.
- Backlash in the table feed screw is less troublesome than in down milling.
- Surface finish is generally poorer than down milling because the tooth rubs before taking a definite chip.

2.2 Down milling (climb milling)

- Cutter rotates in the same direction as the work feed at the point of contact.
- Chip thickness starts at a maximum and reduces approximately to zero.
- Cutting force tends to press the work against the table.
- Usually gives better surface finish and lower rubbing.
- Requires a rigid machine and backlash-elimination arrangement because the cutter tends to pull the work into the cut.

Feature	Up milling	Down milling
Cutter rotation relative to feed	Opposite direction	Same direction
Chip thickness	Zero to maximum	Maximum to zero
Work-force tendency	Lifts work	Presses work down
Backlash sensitivity	Lower	Higher
Surface finish	Generally less smooth	Generally better

3. Milling Operations

Operation	Purpose
Plain / slab milling	Produces a flat surface parallel to cutter axis using a peripheral cutter.
Face milling	Produces a flat surface approximately perpendicular to cutter axis.
End milling	Uses an end mill; cutting occurs at end and periphery for slots, pockets and profiles.
Side milling	Uses side teeth to machine vertical surfaces or slots.
Straddle milling	Two side cutters machine two parallel vertical surfaces simultaneously.
Gang milling	Two or more cutters mounted on one arbor perform several surfaces in one pass.
Angular milling	Produces surfaces at an angle using single- or double-angle cutters.
Form milling	Produces irregular profiles using a cutter having the required form.
Slot / keyway milling	Produces straight slots and keyways using slotting or end-milling cutters.
T-slot milling	Produces T-shaped slots, usually after first making a straight slot.
Dovetail milling	Produces dovetail guideways or slots using a dovetail cutter.
Helical milling	Produces helical grooves, flutes and similar surfaces using coordinated table movement.
Gear milling	Cuts gear-tooth spaces using a form-relieved gear cutter and indexing.
Thread milling	Produces threads using a suitable thread milling cutter.
Profile / contour milling	Machines curved or irregular outlines, especially on vertical/CNC milling machines.

4. Milling Cutters

Cutter	Application / characteristic
Plain/slab cutter	Teeth on periphery; used for machining broad flat surfaces.
Side milling cutter	Teeth on periphery and sides; used for slots and side surfaces.
Side-and-face cutter	Heavy-duty slot cutter with side and peripheral cutting edges.
Face milling cutter	Cutting edges mainly on face and periphery; machines large flat surfaces.

End mill	Cuts at end and periphery; used for slots, pockets, profiles and contouring.
Slot drill	End-mill type cutter capable of plunging and slot cutting.
Angle cutter	Single- or double-angle cutter for V-grooves, angular surfaces and serrations.
Form cutter	Profiled cutter for concave, convex or special shapes.
T-slot cutter	Used after a pilot slot to generate the undercut T-slot.
Woodruff keyseat cutter	Small disc-type cutter for Woodruff keyways.
Gear cutter	Form-relieved involute cutter selected according to module/DP and tooth number.
Fly cutter	Single-point cutter mounted in a rotating holder; used for broad flat finishing cuts.

4.1 Cutter nomenclature and selection factors

- Cutter diameter, width, bore and number of teeth.
- Helix angle and rake/clearance geometry.
- Cutter material such as HSS or carbide.
- Required operation, work material, rigidity and production rate.
- For gear cutters, module or diametral pitch and tooth-number range must match the gear to be cut.

5. Indexing and Indexing Methods

Indexing is the operation of rotating a workpiece through a definite angular amount so that equally spaced features such as gear teeth, splines, flutes or polygonal faces can be machined.

5.1 Dividing head principle

In a standard dividing head, the worm and worm wheel commonly have a ratio of 40:1. Thus 40 turns of the index crank rotate the spindle by one complete revolution.

$$\textbf{Crank movement per division} = 40 / N$$

where N is the number of equal divisions required.

5.2 Direct indexing

In direct indexing, the worm is disengaged and the spindle is indexed directly using a plate attached to the spindle. If the plate has 24 holes, the possible divisions are those obtained directly from the 24-hole circle.

$$\textbf{Holes moved per division} = \text{Number of holes in direct plate} / N$$

5.3 Simple (plain) indexing

The worm remains engaged. The crank is turned by the fraction 40/N for every division. The fractional part is converted into a convenient number of holes on one of the available index circles.

$$\textbf{Crank turns per division} = 40 / N$$

5.4 Compound indexing

Compound indexing is used when the required fraction cannot be obtained on a single available hole circle. The crank is moved through two fractional movements using two hole circles in the same or opposite directions.

$$\textbf{Required crank movement} = A/a \pm B/b$$

Here A and B are numbers of holes moved on hole circles having a and b holes respectively. The combination is selected so that the total equals 40/N.

5.5 Differential indexing

Differential indexing is used for numbers of divisions that cannot be obtained by simple indexing. A nearby convenient number N' is selected for simple indexing, while the index plate is made to rotate slowly through a change-gear train as the crank is turned.

$$\textbf{Approximate crank movement} = 40 / N'$$

$$\textbf{Change-gear ratio} = 40(N' - N) / N'$$

The sign determines whether the index plate rotates in the same or opposite direction to the crank. The actual gear arrangement depends on the dividing-head setup.

5.6 Angular indexing

For an angular movement θ in degrees, a 40:1 dividing head requires:

$$\textbf{Crank movement} = 40\theta / 360 = \theta / 9 \text{ turns}$$

Method	When used	Main relation
Direct indexing	Simple divisions possible directly on spindle plate	Holes moved = H/N
Simple indexing	Most ordinary divisions	40/N crank turns
Compound indexing	Fraction not possible on one hole circle	A/a ± B/b
Differential indexing	Difficult/non-obtainable division by simple indexing	Use N' plus change gearing
Angular indexing	Specified angle rather than equal divisions	θ/9 crank turns

6. Shaper Mechanism

A shaper machines surfaces with a reciprocating single-point cutting tool carried by a ram. The work is normally stationary during a cutting stroke and receives a small feed at the end of each cycle.

6.1 Crank and slotted-link quick-return mechanism

- The bull gear is driven at approximately uniform speed.
- A crank pin on the bull gear moves a sliding block in the slotted rocker arm.
- Oscillation of the rocker arm is transmitted to the ram, producing reciprocation.
- The crank rotates through a larger angle during the cutting stroke and a smaller angle during the return stroke.
- Therefore the return stroke takes less time than the cutting stroke - the quick-return effect.

6.2 Hydraulic shaper mechanism

- A hydraulic cylinder drives the ram directly.
- Flow-control and reversing valves control cutting and return strokes.
- Cutting speed can be made nearly uniform over most of the stroke.
- The system provides smoother motion and easier speed adjustment than a mechanical quick-return drive.

6.3 Feed mechanism

A ratchet-and-pawl or equivalent arrangement gives an intermittent feed to the table, generally near the end of the return stroke. The feed may be horizontal, vertical or angular depending on the required surface.

7. Shaper Operations

Operation	Description
Horizontal surface	Tool reciprocates horizontally while the table receives cross feed.
Vertical surface	Tool head/swivel arrangement gives vertical feed to machine a vertical face.
Angular surface	Tool head or table is set to the required angle.
Step / shoulder	Combination of horizontal and vertical surfaces.

Slot and groove	Narrow tool machines slots, keyways and rectangular grooves.
Dovetail	Angular cutting tool or angular setting produces dovetail surfaces.
Irregular / contoured surface	Form tool or controlled feed arrangement follows the required contour.
Keyway	Internal/external keyways can be cut with suitable setup.

8. Planer Mechanism

A planer is used for large and heavy workpieces. The work is mounted on a reciprocating table, while one or more cutting tools are held in tool heads mounted on the cross-rail and/or columns.

8.1 Table drive and reversing mechanism

- The table reciprocates on long accurately machined bed guideways.
- Older planers use mechanical reversing drives such as gear/rack or belt arrangements; modern planers may use hydraulic or electric drives.
- The cutting stroke is comparatively slower and the return stroke is faster to reduce idle time.
- At the end of each stroke, the tool head receives a small feed for the next cut.

8.2 Cross-rail and tool-head mechanism

- The cross-rail is vertically adjustable to accommodate different work heights.
- Tool heads can move horizontally along the cross-rail.
- Side tool heads, where provided, machine vertical side surfaces.
- Several tools may cut simultaneously, increasing productivity on large jobs.

9. Planer Operations

Operation	Description
Large horizontal surfaces	Machining machine beds, plates and long flat surfaces.
Vertical surfaces	Side/rail tool heads machine vertical faces.
Angular surfaces	Tool head is swivelled or work is set at the required angle.
Steps and shoulders	Produced by coordinated setting of tool heads.
Slots and grooves	Straight slots and grooves can be machined over long lengths.
Guideways	Planing of V-ways, flat ways and machine-tool bed guide surfaces.
Multiple surfaces	Several tool heads can machine different surfaces in one table stroke.

10. Cutting Speed, Feed, Depth of Cut and Machining Time - Shaper & Planer

10.1 Cutting speed

Cutting speed is the average linear speed of the tool relative to the work during the cutting stroke. Because mechanical shapers and planers use quick return, cutting and return times are unequal.

If N = complete cycles/min and Q = t_c / t_r, then $t_c = [Q/(Q + 1)](1/N)$ min

Average cutting speed $V = L N (Q + 1) / (1000 Q)$ m/min

where L is the cutting-stroke length in mm, t_c is cutting-stroke time, t_r is return-stroke time, and Q is the quick-return ratio t_c/t_r (>1). If the number of cutting strokes per minute is given directly and return time is not being considered, use the stated cutting-stroke rate consistently.

10.2 Feed

Feed is the distance by which the work or tool is advanced after each cutting stroke. For shapers it is commonly stated in mm per stroke; for planers it is the tool-head feed per table stroke.

Feed rate across surface = $f \times N$ mm/min

where f = feed per cutting stroke (mm/stroke) and N = cutting strokes/cycles per minute as defined in the problem.

10.3 Depth of cut

Depth of cut is the thickness of material removed in one pass, measured perpendicular to the machined surface. It is set by vertical adjustment of the tool or table/tool head.

10.4 Machining time

If a surface of effective width W is machined with feed f per cutting stroke, the number of cutting strokes required is:

$$\textbf{Number of cutting strokes} = W / f$$

$$\textbf{Machining time } T = W / (fN) \text{ min}$$

W should include approach and over-travel/feed allowances where specified. For multiple depth passes, multiply the time per pass by the required number of passes.

10.5 Number of passes

$$\textbf{Number of depth passes} = \text{Total stock to be removed} / \text{Depth of cut per pass}$$

Round upward to the next whole pass where necessary.

11. Shaper vs Planer

Point	Shaper	Planer
Reciprocating member	Tool/ram reciprocates	Work/table reciprocates
Work size	Small and medium work	Large and heavy work
Tool heads	Usually one	One or several
Feed	Usually given to work/table	Usually given to tool heads
Cutting capacity	Light to moderate cuts	Heavy cuts
Machine size	Compact	Large and massive
Power requirement	Lower	Higher
Productivity	Lower for large surfaces	Higher on large jobs; simultaneous tools possible
Quick return	Commonly crank/slotted-link or hydraulic	Table reversing drive provides faster return
Typical application	Slots, keyways, small flat/angular surfaces	Machine beds, long guideways, large plates

12. Simple Numerical Problems

Problem 1: Simple indexing

Find the crank movement per division for cutting 32 equally spaced slots with a 40:1 dividing head.

$$\textbf{Crank movement} = 40/32 = 5/4 = 1\ 1/4 \text{ turns per division}$$

Hence rotate the crank 1 full turn plus 1/4 turn for each division.

Problem 2: Angular indexing

Find the crank movement required to rotate the work through 27° on a 40:1 dividing head.

$$\textbf{Crank movement} = \theta/9 = 27/9 = 3 \text{ turns}$$

Problem 3: Milling spindle speed

A 100 mm diameter milling cutter operates at a cutting speed of 25 m/min. Find spindle speed.

$$N = 1000V/(\pi D)$$

$$N = (1000 \times 25)/(\pi \times 100) = 79.6 \text{ rpm} \approx 80 \text{ rpm}$$

Problem 4: Milling table feed

A cutter has 8 teeth and rotates at 250 rpm. If feed per tooth is 0.08 mm, find table feed.

$$\textbf{Table feed } F = f_z \times Z \times N$$

$$F = 0.08 \times 8 \times 250 = 160 \text{ mm/min}$$

Problem 5: Milling time

A milling operation requires a total table travel of 180 mm. The cutter has 10 teeth, feed per tooth is 0.06 mm and spindle speed is 300 rpm. Find machining time.

$$\text{Feed rate} = 0.06 \times 10 \times 300 = 180 \text{ mm/min}$$

$$T = 180/180 = 1.00 \text{ min}$$

Problem 6: Shaper cutting speed with quick return

A shaper has a cutting stroke length of 400 mm and operates at 30 cycles/min. The quick-return ratio t_c/t_r is 3/2. Find the average cutting speed.

$$V = L\,N\,(Q + 1)/(1000Q)$$

$$V = 400 \times 30 \times (1.5 + 1)/(1000 \times 1.5) = 20.0 \text{ m/min}$$

Problem 7: Shaper machining time

A surface 120 mm wide is to be machined on a shaper at a feed of 0.5 mm/stroke. The machine runs at 24 cycles/min. Neglect allowance. Find machining time for one pass.

$$\text{Number of cutting strokes} = W/f = 120/0.5 = 240$$

$$T = W/(fN) = 120/(0.5 \times 24) = 10.0 \text{ min}$$

Problem 8: Planer machining time with allowance

A planer is to machine a surface 300 mm wide. Feed is 1.0 mm per cutting stroke, and the table completes 12 cycles/min. Allow 10 mm total feed allowance. Find time for one pass.

$$\text{Effective feed width } W = 300 + 10 = 310 \text{ mm}$$

$$T = 310/(1.0 \times 12) = 25.83 \text{ min}$$

Problem 9: Number of depth passes and total time

A shaper must remove 6 mm thickness from a surface in cuts of 2 mm depth. The machining time per pass is 8 min. Find total number of passes and total cutting time.

$$\text{Number of passes} = 6/2 = 3$$

$$\text{Total machining time} = 3 \times 8 = 24 \text{ min}$$

13. Important Points

- In milling, the cutter rotates continuously while the work is fed relative to it.
- Up milling has chip thickness increasing from approximately zero; down milling has chip thickness decreasing from maximum to zero.
- For a standard 40:1 dividing head, simple indexing crank movement is 40/N turns per division.
- Direct, simple, compound, differential and angular indexing are selected according to the required division.
- A shaper reciprocates the cutting tool; a planer reciprocates the work.
- The quick-return mechanism reduces idle return time.
- Shapers are suited to small/medium jobs; planers are suited to large and heavy jobs.

- Feed for shaper/planer is usually expressed in mm per cutting stroke.
- Machining time for one surface pass is $T = W/(fN)$, provided N and f are defined consistently.
- Depth of cut is measured perpendicular to the machined surface.

Typical Examination Questions

1. Explain the working mechanism of a milling machine.
2. Differentiate between up milling and down milling.
3. Explain the important milling operations and milling cutters.
4. Define indexing and explain direct, simple, compound and differential indexing.
5. Explain the crank and slotted-link quick-return mechanism of a shaper.
6. Explain the mechanism and operations of a planer.
7. Differentiate between a shaper and a planer.
8. Define cutting speed, feed and depth of cut for a shaper/planer.
9. Derive the machining-time relation for shaper and planer operations.
10. Solve numerical problems on indexing, milling feed/time and shaper/planer machining time.

Reference

Hajra Choudhury, S. K. and Hajra Choudhury, A. K., Elements of Workshop Technology, Vol. II.